

② 8 pt. DIT FFT flowgraph

$$8\text{pt } x(n) = \{x(0), x(1), x(2), x(3), x(4), x(5), x(6), x(7)\}$$

$$4\text{pt } \begin{cases} x_e(n) = x(0), x(2), x(4), x(6) & N=4 \\ x_o(n) = x(1), x(3), x(5), x(7) & N=4 \end{cases}$$

$$2\text{pt } \begin{cases} x_1(n) = \{x(0), x(4)\} \\ \{x(2), x(6)\} \\ \{x(1), x(5)\} \\ \{x(3), x(7)\} \end{cases}$$

$$W_8^0 = 1$$

$$W_8^1 = 0.707 - j0.707$$

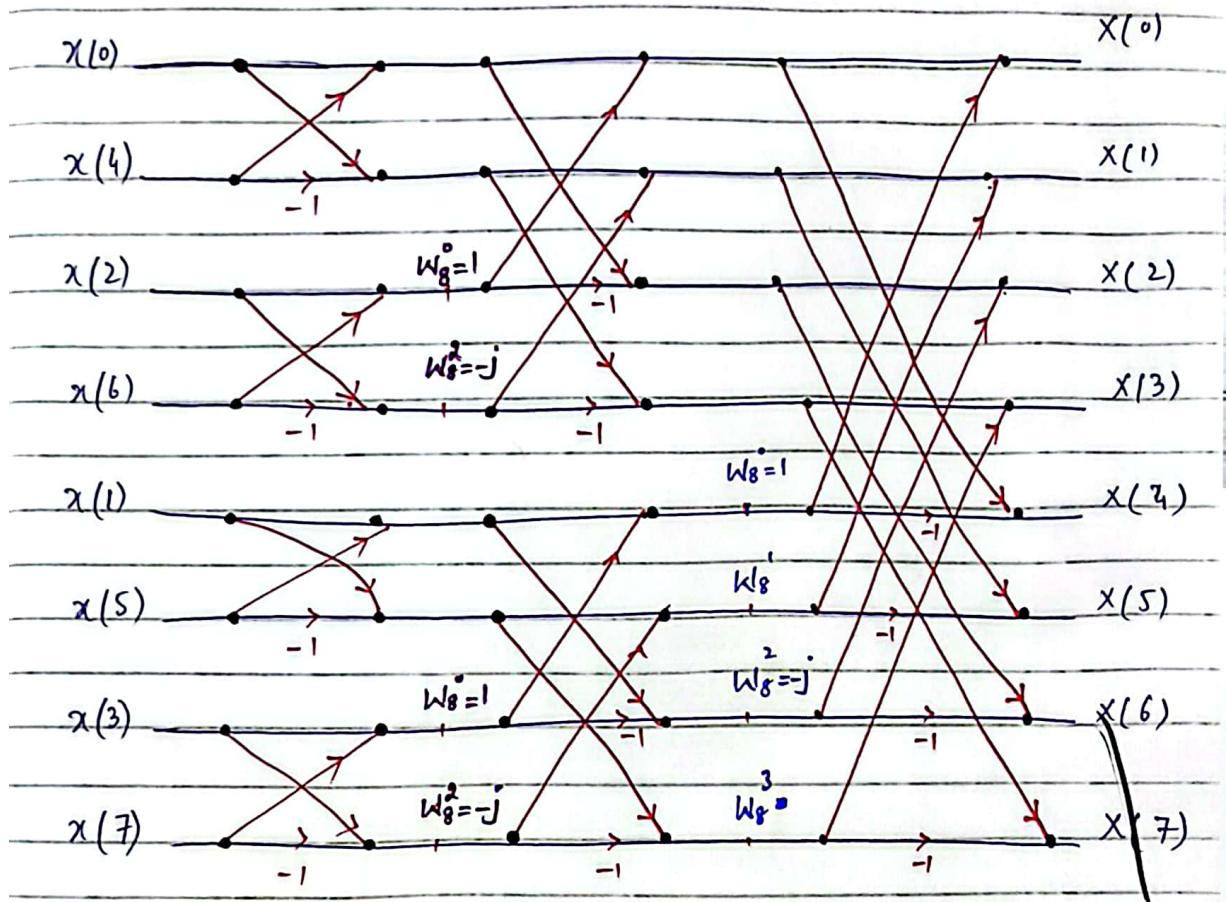
$$W_8^2 = -j$$

$$W_8^3 = -0.707 - j0.707$$

$$e^{-j\frac{2\pi}{8}k}$$

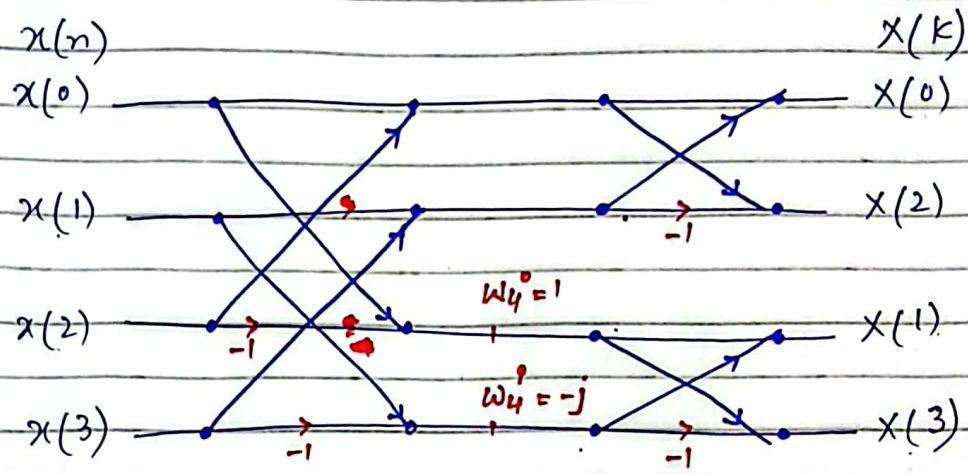
2 5

8 pt. DIT FFT

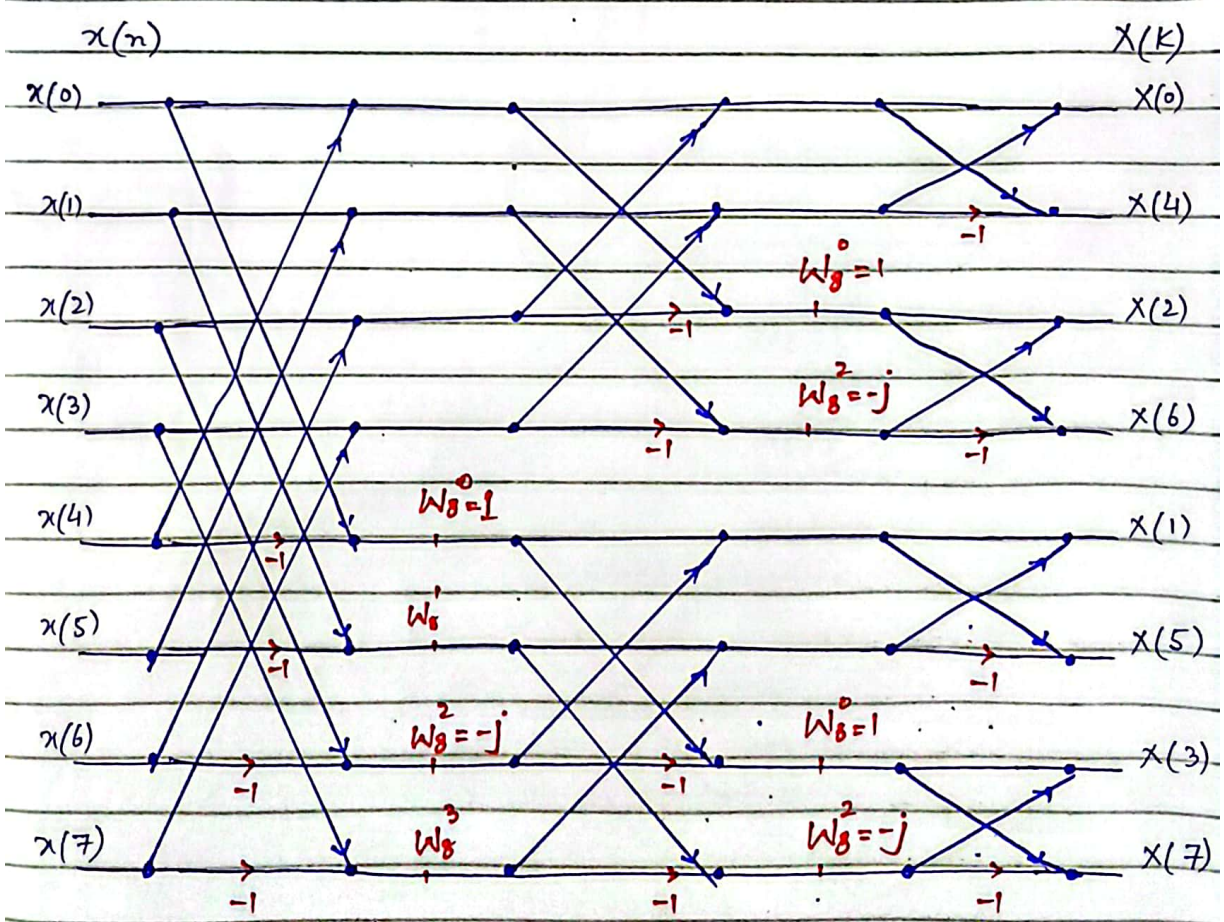


3

③ 4 pt. DIF FFT

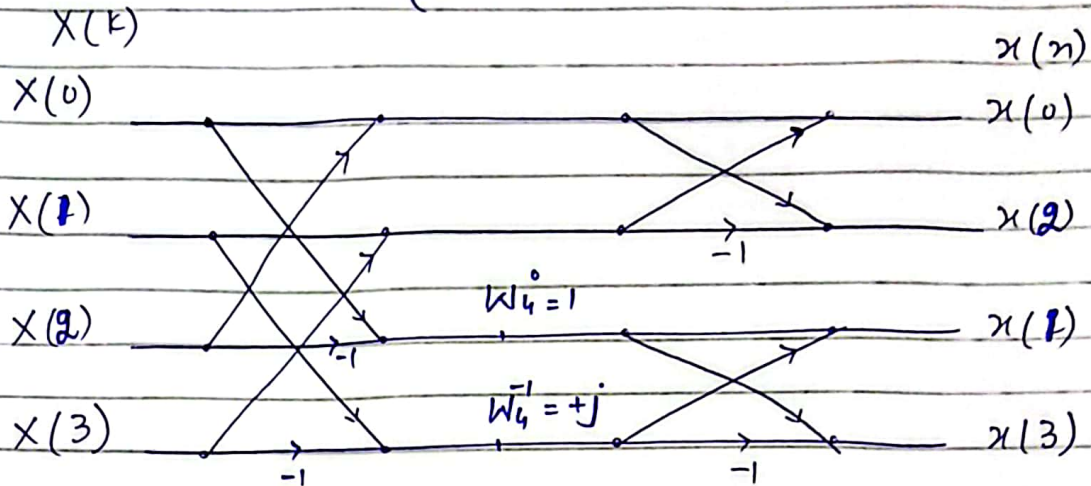


④ 8 pt. DIF FFT



~~4~~ DIT $\rightarrow$ IFFT ($X(k) \rightarrow x(n)$) (4)

(5) $N=4$ pt DIT IFFT
(Decimation in time)



here Twiddle factor is complex conjugate

(6) $N=8$ pt. DIT-IFFT

